

How do RNNs encode mixtures? One mechanism for three geometries

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Abstract

Artificial and biological networks encode inputs as activity patterns distributed across many units. Naturalistic inputs are often superimpositions of several simpler components, as overlapping gratings in vision, mixed voices in audition, odor blends in olfaction. For such *mixtures*, neurophysiology reports three response geometries: either both component responses are combined in proportion to their input strength, or one is favored over the other, or a new pattern emerges that neither component alone elicits (Busse et al., 2009; Rust et al., 2006; Zoccolan et al., 2005). Which circuit properties determine which encoding arises? The divisive normalization model (Carandini & Heeger, 2012) captures the phenomenology of the first two geometries, but does not identify the underlying network mechanism, and cannot express emergence. This theoretical study analyzes a recurrent-network model and derives, analytically, the circuit property that selects each geometry. The central result is that a *single* network can produce all three geometries as the input varies, with no change in synaptic weights. This flexibility has computational implications for downstream readouts.

Framework: response geometries in RNNs Parametrizing mixture responses

In distributed coding, a *mixture input* corresponds to a linear combination of two single inputs: $\vec{u}_{\text{mix}} = \alpha_1 \vec{u}_1 + \alpha_2 \vec{u}_2$, with weights $\alpha_1, \alpha_2 \geq 0$. A network maps each input to a population response, a vector in activity space: the *component responses* $\vec{r}_1, \vec{r}_2$ for each single input; the *mixture response* $\vec{r}_{\text{mix}}$ for their combination. To describe the geometry of the mixture response, the latter can be split into a part in the components' plane and an orthogonal residual (Fig. 1(D)): $\vec{r}_{\text{mix}} = \beta_1 \vec{r}_1 + \beta_2 \vec{r}_2 + \beta_{\perp} \vec{\eta}_{\perp}$. The coefficients classify the three geometries. The *unbiased combination* is the proportional sum that matches the input ($\vec{r}_{\text{ref}} \propto \alpha_1 \vec{r}_1 + \alpha_2 \vec{r}_2$). The *biased combination* has a different in-plane balance ($\beta_1 : \beta_2 \neq \alpha_1 : \alpha_2$). The *emergent pattern* is characterized by a non-vanishing orthogonal magnitude ($\beta_{\perp} \neq 0$). The goal is to explain how the coefficients derive from circuit mechanisms.

From the circuit to intrinsic coordinates

Computing the coefficients requires a circuit that generates responses. A tractable model is a recurrent network

of N neurons whose connectivity decomposes as a sum of R interaction patterns, $\mathbf{J} = \frac{1}{N} \sum_{r=1}^R \mathbf{m}_r \mathbf{n}_r^{\top}$ (Fig. 1(A)). This structure is observed in cortical connectomes (Clark et al., 2025) and in RNNs trained on cognitive tasks (Dubreuil et al., 2022; Mastrogiuseppe & Ostojic, 2018; Schuessler et al., 2020a, 2020b). Under this parametrization, the responses of all neurons can be reduced to a lower-dimensional vector of R collective coordinates. The network maps each input $\vec{u}$ to a response $\vec{r}$ through:

$$\vec{r} = g \left(\mathbf{I} - g \mathbf{\Sigma}^{(nm)} \right)^{-1} \mathbf{\Sigma}^{(nl)} \vec{u} =: M(g) \vec{u} \quad (1)$$

The map factorizes into two stages (Fig. 1(B)). First, the fixed feedforward matrix $\mathbf{\Sigma}^{(nl)}$ projects the input into the collective variables, setting *which directions the stimulus drives*. Then, recurrent feedback amplifies each direction via the matrix $\mathbf{\Sigma}^{(nm)}$ — that encodes recurrent patterns — and a scalar *gain* g — that measures the global response sensitivity at the current activity level. The gain itself depends on the average input received by neurons via the activation function. Both external inputs and recurrent activity contribute to the total input variance Δ that affects the gain, via $\Delta = \Delta_0 + \vec{u}^{\top} \mathbf{C}_u \vec{u} + \vec{r}^{\top} \mathbf{C}_r \vec{r}$, where the matrices $\mathbf{C}_u, \mathbf{C}_r$ specify pooled variances.

Once the response is expressed for any input, describing *mixture* responses requires a fixed coordinate system. The recurrent matrix $\mathbf{\Sigma}^{(nm)}$ provides such an intrinsic basis: its eigenvectors $\vec{v}$ set the *principal directions* of the network's activity. Along each principal direction k , the response coordinate writes $z_k = A_k(g) \xi_k$. The *drive* $\xi_k = \vec{v}_k^{\top} \mathbf{\Sigma}^{(nl)} \vec{u}$ measures how strongly the input projects onto direction k . The *amplification factor* $A_k(g) = g / (1 - g \lambda_k)$ rescales that drive, depending on the eigenvalue λ_k and the gain g (Fig. 1(C), bottom). The connectivity fixes the eigenvalues/vectors — which directions exist, how steeply each amplifies — while the input and pooling matrices set the gain.

Mechanism: one circuit, three geometries Amplification profile as control quantity

The analysis starts by expressing the mixture response in the intrinsic basis. Along each principal direction, the mixture drive is the linear projection, and is amplified at the mixture gain: $z_{k,\text{mix}} = A_k(g_{\text{mix}})(\alpha_1 \xi_{k,1} + \alpha_2 \xi_{k,2})$. The



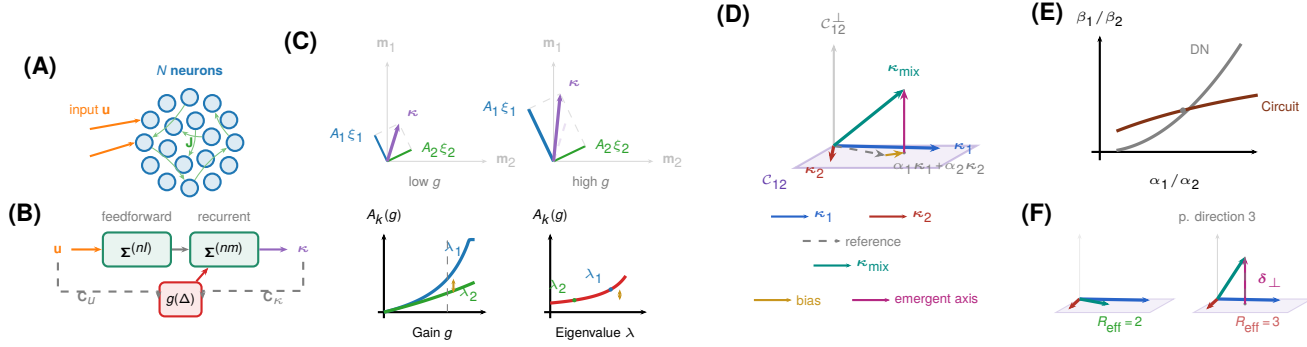


Figure 1: **(A)** Recurrent network of N neurons with connectivity $\mathbf{J}$. **(B)** Effective circuit. The map factorizes into a feedforward stage $\Sigma^{(nl)}$ and a recurrent stage $\Sigma^{(nm)}$, the latter modulated by the gain $g(\Delta)$. The gain depends on the total input via the variance Δ , to which external inputs and recurrent activity contribute through the pooling matrices $\mathbf{C}_u$ and $\mathbf{C}_r$ (dashed). **(C)** *Top*: Higher gain amplifies the dominant principal direction more, rotating the response κ . *Bottom*: Amplification factors depend on the gain (left) and eigenvalue (right). **(D)** The mixture response decomposes into the component span (plane), with a possible bias, and an out-of-span departure. **(E)** The circuit-derived weight ratio saturates at a finite bias, unlike the divisive normalization power law. **(F)** With more than two principal directions, the mixture generically exits the span.

reference unbiased response (assuming both single components elicit the same gain g_c) amplifies the same drive as: $z_{k,\text{ref}} = A_k(g_c)(\alpha_1 \xi_{k,1} + \alpha_2 \xi_{k,2})$. The reference and the mixture therefore differ only in the *gain* at which the shared drive is amplified. The *amplification ratio* isolates that difference along each direction: $r_k = A_k(g_{\text{mix}})/A_k(g_c)$. Its profile across the active principal directions is the decisive quantity. The next step associates each geometry to one profile, then identifies the conditions that create it.

Non-trivial geometries are generic

The unbiased geometry corresponds to a *uniform* amplification across all principal directions: each coordinate is multiplied by the same factor, so the mixture aligns with that reference. Otherwise, coordinates stretch unevenly, and the response rotates (Fig. 1(C), top).

A *non-uniform amplification profile* requires two ingredients, visible in the amplification factor $A_k(g) = g/(1 - g\lambda_k)$: the mixture must change the gain relative to the components, and at least two active principal directions must carry distinct eigenvalues. Both ingredients typically hold in recurrent circuits, so the unbiased geometry is degenerate. First, heterogeneous eigenvalues are expected, unless the connectivity has a fine-tuned symmetry. Second, the gain also generically shifts, because superposing two inputs does not merely add their variances. For convex weights, the mixture input variance is lower than the component average by a term proportional to the input dissimilarity, and a saturating nonlinearity converts this variance change into a gain shift, $g_{\text{mix}} \neq g_c$.

Emergent patterns are also generic. The mixture stays in the plane only if $\vec{\kappa}_{\text{mix}} = \beta_1 \vec{\kappa}_1 + \beta_2 \vec{\kappa}_2$, which must hold coordinate by coordinate. Thus, the two coefficients β_1, β_2 must satisfy one scalar equation per active direction. Two

active directions give two equations: a solution exists, so the mixture stays in the plane. However, each further active direction adds one equation that the two in-plane unknowns cannot generally satisfy, so the mixture acquires an out-of-plane part. Therefore, an emergent pattern is expected as soon as three principal directions are driven by the two single inputs (Fig. 1(F)).

Contrast with divisive normalization

The divisive normalization model expresses the response as a *weighted sum of the two components*, with coefficients $\beta_i^{\text{DN}} = \alpha_i^n / (\sigma^n + \alpha_1^n + \alpha_2^n)$ for $i \in \{1, 2\}$, where σ and n are free parameters. Thus, by construction, this model constrains the response to the component span and cannot describe an emergent pattern. Furthermore, the bias depends on input coefficients through a power law, $\beta_1^{\text{DN}}/\beta_2^{\text{DN}} = (\alpha_1/\alpha_2)^n$. In contrast, the recurrent circuit exhibits an *input-dependent* exponent. In a minimal circuit, the exponent counterpart is the ratio between the amplification ratios of the two principal directions: $n_{\text{span}} = r_-/r_+$. Near balanced input strengths, both models agree, but as stimulus strength grows, the circuit bias saturates, while the power law keeps growing (Fig. 1(E)).

Implications and predictions

A single recurrent circuit can generate all three geometries as the input varies, and the proportional combination is exceptional. For downstream readouts, a biased combination may implement attentional selection by gain control, while an emergent pattern makes the mixture linearly separable from its components and readable as a distinct identity. The model yields concrete predictions: competition should decrease with stimulus intensity, and emergent patterns should depend on input similarity.

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